PY103 · Object-Oriented Programming (OOP) in Python
Clock Hours: 50
Quarter Credit Hours: 4
Prerequisites: PY102
Instructors
Clifford Osburn
A.A.S. in CIS Software Systems & Networking, Amarillo College - 25 yrs in software engineering.
David Gross
M.S. in Computer Engineering, University of Central Florida - 20 yrs in software engineering.
Douglas Laney
B.S. in Software Engineering & Business Administration, University of Illinois at Urbana-Champaign - 31 yrs in software engineering.
John Berta
12 yrs in software engineering: AP Computer Science instructor since 2014 (Hazleton Area SD); Python engineer at BrickSeek (2018-24); UPenn cert. in Applied Computational Thinking. No degree
in field; holds M.A. in Classroom Technology, Wilkes University.
Neil Kalinowski
M.S. in Computer Science, University of Texas at San Antonio - 26 yrs in software engineering.
Sean Starkey
M.S. in Computer Engineering, Colorado Technical University - 30 yrs in software engineering.
Shiv Hari
M.S. in Software Engineering, George Mason University - 15 yrs in software engineering.
Have an academic question? Submit an Academic Assistance Request, and Student Services will make sure it reaches the right course instructor or member of the academic team.

Course Description
Object-Oriented Programming (OOP) in Python introduces students to programming using Python. Students will explore topics such as What is Object-Oriented Programming?, History and Motivation
for OOP, Benefits and Drawbacks of OOP, Defining and Creating Classes in Python, Instantiating Objects from Classes.

Course Objectives
Object-oriented programming and its motivation
Defining classes and instantiating objects
Constructors, attributes, and instance methods

Encapsulation and object state
Inheritance and polymorphism
Course Outline
Unit Topics & Assignments
Unit 1 – Thinking in objects Topics:
• Object-oriented programming and Python objects
• Object state, data attributes, behavior, and methods
• Classes as blueprints and CapitalCase naming
• Built-in modules, object instances, methods, and generated data
• Minimal class definitions and object instantiation

Assessment:
Weekly review (quiz)
Unit 2 – Bringing objects to life Topics:
• Multiple instances, independent attributes, identity, and equality
• Instance methods, self, and attribute access
• init constructors, parameterized attributes, validation, and safe defaults
• State-reading and state-changing methods with persistent updates
• Instance attributes versus shared class attributes

Assessment:
Weekly review (quiz)
Unit 3 – When objects meet Topics:
• Managing and comparing independent object states
• One-to-one and one-to-many object relationships
• Collection processing through object iteration and methods
• Loop-based object creation from user input
• Logic and state ownership with object invariants

Assessment:
Weekly review (quiz)
Unit 4 – From simple objects to flexible systems Topics:
• Encapsulation with internal attributes and getters
• Setter validation and protected object state
• Subclass inheritance, method overriding, and implementation selection
• Composition versus inheritance in object models
• Polymorphism, str, repr, and readable object output

Assessment:
Final course review (quiz)
Review format
Reviews in this course consist of a MIX of open-ended and

multiple-choice questions, focused on practical understanding of core concepts: applying key ideas, interpreting code, identifying errors, and reasoning about system behavior.

NOTE: The outline is subject to change at the institution’s discretion

Text & Reference Materials
Proprietary instructional content
Python Docs – OOP: https://docs.python.org/3/tutorial/classes.html
Teaching Strategies
Individualized tutorials and demonstrations.
LMS Assignments.
Hands-on practice labs.
Projects.
Interactive discussions and case studies.
Grading Policies
Students will be evaluated in the following manner:

Quizzes (Weekly Reviews): 30%.
Online Assignments (Discussions): 10%.
Tests/Exams (Final Review): 60%.
Bonus: Students who submit and pass all reviews on time are eligible to earn up to 10 bonus points.

The grading percentages noted may change at the discretion of the institution.

To satisfactorily complete this course, a student must achieve a minimum of 60 percent (a grade of D). In order to graduate, a student must attain a minimum cumulative GPA of 2.0 and complete all courses with a passing grade.

Grading Scale
Final grades are based on the final grading system found in the catalog.

Grade	Numeric Grades	Grade Points
A	100 – 90	4.0
B	89 – 80	3.0
C	79 – 70	2.0
D	69 – 60	1.0
F	59 & below	0.0

Reviews

All reviews must be completed by the end of the term for course completion. At term end, grades finalize and reviews lock – the only exception is the final review retake, open for two weeks a
fter the last learning day.

Weekly Reviews
A formative assessment that helps students reinforce each unit’s content.

Weekly reviews are scored, but only pass/fail affects your grade: a result of D or higher counts as a Pass and earns full points for that review; a higher passing grade doesn’t add extra
course-grade points.
A result below D counts as a Fail and earns no points. A failed attempt does not complete the lesson, and you may try again.
Weekly reviews can be retaken as long as the course remains open for learning.
Weekly reviews are graded automatically, and results are available immediately.
Your first passing attempt completes the lesson and allows you to continue.
Discussions
A weekly question or exercise aligned with the content covered in the current academic unit and the cohort’s standard learning pace.

A discussion question is published each week, and students are encouraged to respond on time and in alignment with the material covered in their current academic unit.
All discussion responses may be submitted until the end of the current course.
When a discussion response is submitted, the final course grade is automatically recalculated and updated based on all grade components.
Final Reviews
The exam is graded by an instructor assigned to the course.
Students may take a retake exam; the higher of the two grades is the final exam grade. The retake window is open for two weeks after the last learning day in the term.
Exam grades will be published within 14 days of submission.
The final course grade will be published on the later of the following dates: eight days after the course ends, or the date the exam is graded.
If a student takes the retake exam attempt, the final course

grade will be updated after that exam is graded.
Instructor Availability
This course requires significant individual effort. If you experience any challenges, contact Student Services as soon as possible for academic support or to connect with your course instructor.
Seeking help early can make it easier to stay on track and succeed in the course.

Student Responsibilities
This course is part of Maestro College’s asynchronous distance education program, a non-attendance-taking, academic-event-based program. Coursework is completed through the Maestro Learning
Platform in an asynchronous online format, allowing students to access materials at any time throughout the week. There are no scheduled class meetings; however, students are expected to
complete assignments at least weekly, meet deadlines, and participate regularly in online discussions and activities.

Attendance & Enrollment
Requirements for attendance in asynchronous distance education programs:

Maestro College Asynchronous Distance Education programs are non-attendance taking, academic event based programs. Although these programs are non-attendance taking, the institution is required to document students’ participation in academic related activity and to terminate the enrollment of any student for which this activity cannot be documented at the following points:

at the end of the first calendar week of the academic term;
at the end of the first four calendar weeks of the academic term;
at the midpoint of each academic term; and
at the end of each academic term.
Example: For a program which is run in 10-week terms a student must complete a qualifying academic activity in weeks 1, 4, 5, and 10 to remain active. Failure to complete a qualifying academic activity during any of these weeks will result in the student being dropped from the program.

Samples of qualifying academic events include:

Attending a synchronous class, lecture, recitation, or field

or laboratory activity, physically or online, where there is an opportunity for interaction between the instructor and students
Submitting an academic assignment
Taking an assessment or an exam
Participating in an interactive tutorial, webinar, or other interactive computer-assisted instruction
Participating in a study group, group project, or an online discussion that is assigned by the institution
Interacting with an instructor about academic matters
The following do NOT qualify as academic events:

Logging into an online class or tutorial without any further participation; or
Participating in academic counseling or advisement.
Academic Support
Students enrolled in asynchronous online programs receive academic support through structured, asynchronous support channels aligned with the online instructional model. To request academic assistance, students must submit an academic support ticket through the Maestro Platform, which is administered by Student Services. Academic questions submitted through this system are routed to qualified instructional staff, who provide clarification, guidance, or additional explanation related to course content.

Academic Integrity
You are encouraged to discuss course concepts and assignments with classmates to deepen your understanding, but all submitted work must be your own. Do not copy another person’s work or submit or share solutions that are not your own.

If an academic integrity violation is confirmed, the College may impose sanctions, as determined by the appropriate academic authority and as outlined in the school catalog.

AI Usage
Students may use AI tools for non-graded academic support, such as:

Reviewing material
Asking practice questions
Practicing skills
Students may not:

Use AI to generate graded assignments unless explicitly permitted

Present AI-generated answers as original work without citation
AI misuse may be treated as academic dishonesty.

NOTE: This syllabus is subject to change at the institution’s discretion